PYGEOVISION · GEOSPATIAL ANALYSIS REPORT

Building Footprints

n/a (published dataset) · 98.0 km²

96,788	1,591.6 ha	16.2 %	986.0
Buildings	Total footprint area	Share of the area covered by buildings	Buildings per km²

Executive summary

- 96,788 buildings from Microsoft ML Building Footprints fall inside the area, covering 1,591.6 ha (16.2 % of the area; 986 per km²).
- These are the dataset's polygons, not detections on your own imagery.

> Microsoft Global ML Building Footprints: machine-learning detections on Bing/Maxar/Vexcel imagery of various dates (about 1% false positives reported by the project); building heights where available.

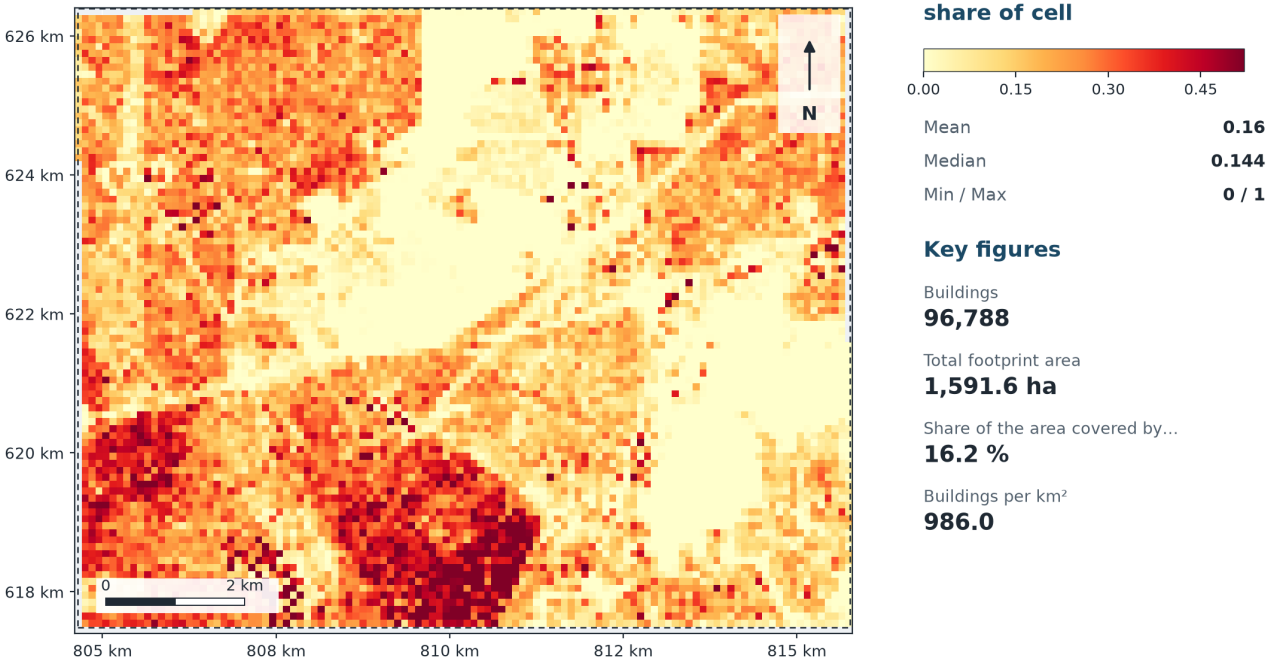
Study area and inputs

Area of interest (west, south, east, north)	-0.2500, 5.5800, -0.1500, 5.6600
Centre	5.6200°, -0.2000°
Area of interest	98.01 km²
Period	n/a (published dataset)
Output grid	112 × 90 pixels, EPSG:32630

Results

Building Footprints

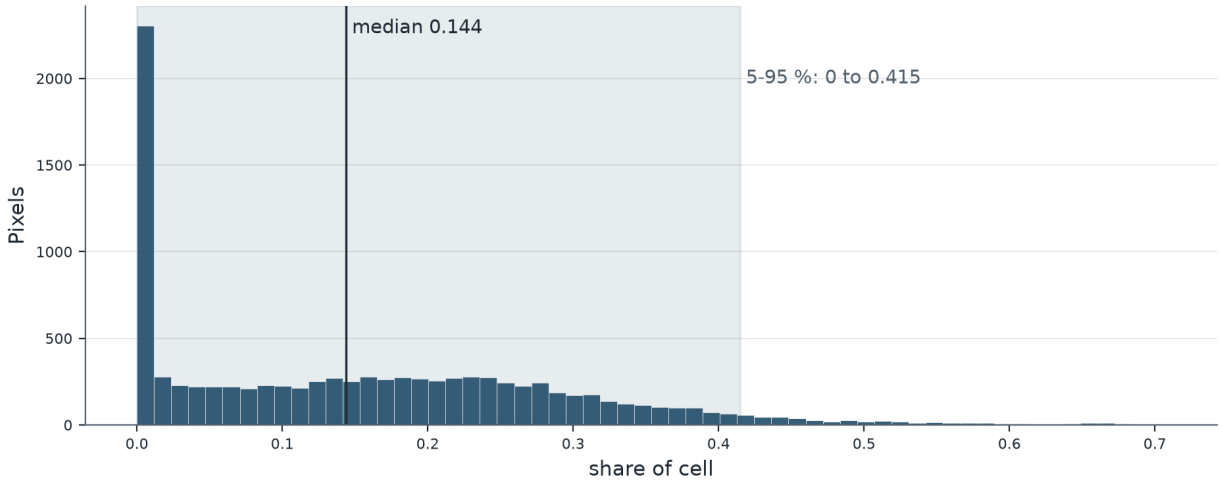
Share of each cell covered by buildings · n/a (published dataset) · AOI -0.2500, 5.5800, -0.1500, 5.6600



CRS: EPSG:32630 · Areas measured from the output raster · No basemap · PyGeoVision 2.2.0 · 2026-10-17 21:32 UTC

Map: Building Footprints

Distribution: Share of each cell covered by buildings



Summary chart

Key statistics (reported by the pipeline)

Measure	Value
Buildings	96,788
Total footprint area	1,591.6 ha
Share of the area covered by buildings	16.2 %
Buildings per km²	986.0
Mean footprint (m²)	164.4
Median footprint (m²)	110.7
Highest cell coverage	100.0 %
Area (km²)	98.17

Other statistics

Measure	Value
Source	microsoft
Cell m	100
Footprints file	accra_buildings/building_footprints.gpkg
Buildings with height	0
Mean height m	n/a

Value statistics: Share of each cell covered by buildings

Statistic	Value
Minimum (share of cell)	0.000
5th percentile (share of cell)	0.000
25th percentile (share of cell)	0.018
Median (share of cell)	0.144
Mean (share of cell)	0.160
75th percentile (share of cell)	0.252
95th percentile (share of cell)	0.415
Maximum (share of cell)	1.000
Standard deviation (share of cell)	0.148

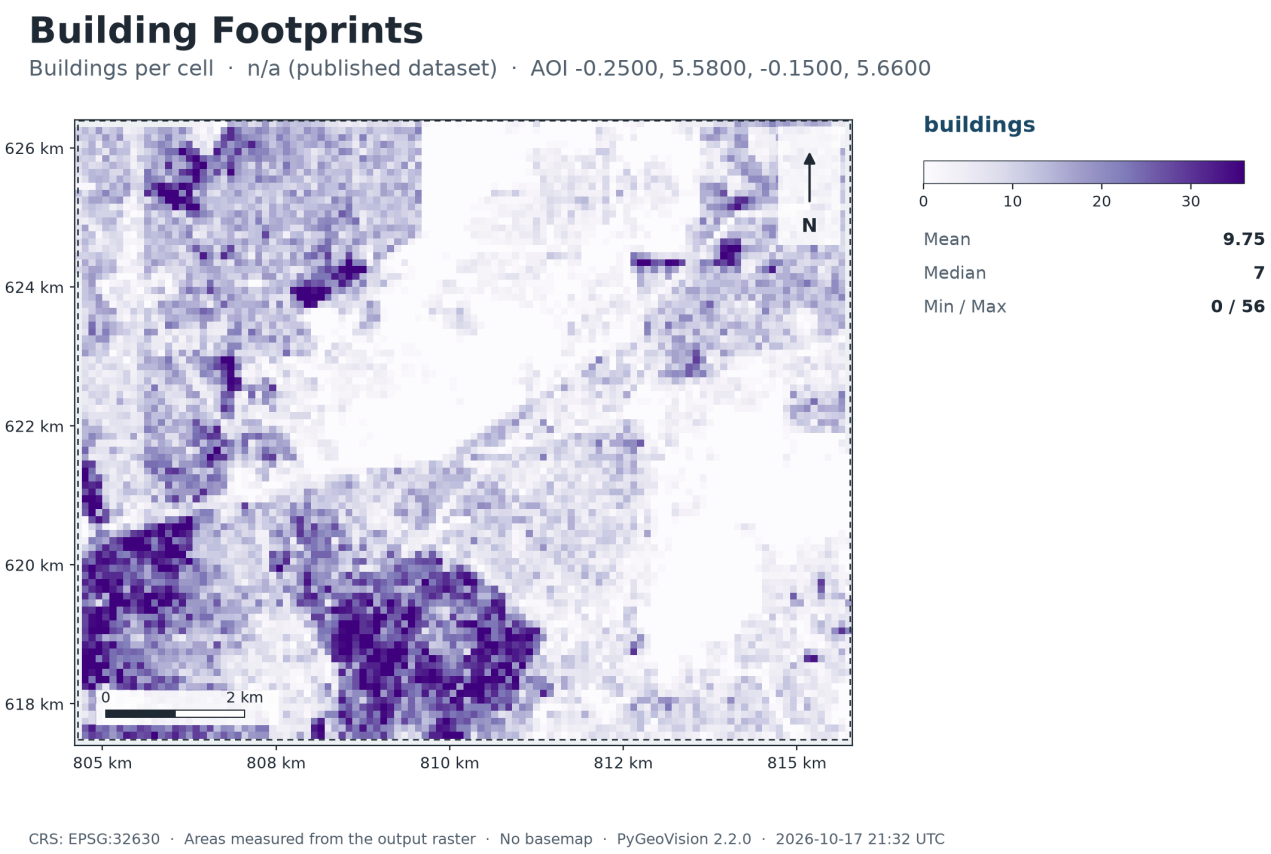
Over 9,836 valid pixels.

Value statistics: Buildings per cell

Statistic	Value
Minimum (buildings)	0.000
5th percentile (buildings)	0.000
25th percentile (buildings)	1.000
Median (buildings)	7.000
Mean (buildings)	9.746
75th percentile (buildings)	15.000
95th percentile (buildings)	30.000
Maximum (buildings)	56.000
Standard deviation (buildings)	9.646

Over 9,836 valid pixels.

Additional layers



Map: Buildings per cell

Method and parameters

Where are the buildings in the area, how many, and how much ground do they cover?

Published footprints (default): the polygons of Microsoft's Global ML Building Footprints, Google's Open Buildings v3 or OpenStreetMap are read for the box, clipped to it and measured in the local UTM zone. Nothing is detected on your imagery.

building_density.tif is the share of each cell covered by building area (a building counts in the cell holding its centroid); building_count.tif counts buildings per cell.

With a checkpoint the pipeline instead runs a segmentation network you trained (class 1 = building) and writes building_mask.tif.

Pipeline	building_footprints
Area of interest	-0.2500, 5.5800, -0.1500, 5.6600
cell_m	100
cloud_mask	True
source	microsoft
period	n/a (published dataset)

Interpretation and limitations

How to read the result

- Dataset footprints differ in date: a recent building may be missing and a demolished one still present.

Limitations

- Published footprints are only as complete, accurate and current as their source; compare the three sources if it matters.
- A building_mask.tif result (only when you pass a checkpoint) is the output of a network you trained and has not been checked against reference data; at 10 m individual houses are not resolvable.

Deliverables

Files

File	Type	Contents
accra_buildings/building_density.tif	GeoTIFF raster	Share of each cell covered by buildings
accra_buildings/building_count.tif	GeoTIFF raster	Buildings per cell
accra_buildings/deliverables/building_footprints_report.pdf	PDF report	Complete analysis report
accra_buildings/deliverables/building_footprints_report.xlsx	Excel workbook	Complete analysis report
accra_buildings/deliverables/building_footprints_report.docx	Word report	Complete analysis report
accra_buildings/deliverables/maps/building_footprints_map.png	PNG image	Map or chart for slides and documents
accra_buildings/deliverables/maps/building_footprints_map_count.png	PNG image	Map or chart for slides and documents
accra_buildings/deliverables/maps/building_footprints_histogram.png	PNG image	Map or chart for slides and documents
accra_buildings/deliverables/vectors/building_footprints.gpkg	GeoPackage (all layers)	Vector features with attributes
accra_buildings/deliverables/vectors/geojson/building_footprints_aoi.geojson	GeoJSON, EPSG:4326	Vector features with attributes
accra_buildings/deliverables/vectors/geojson/building_footprints_buildings.geojson	GeoJSON, EPSG:4326	Vector features with attributes
run_summary.json	JSON	Machine-readable record of this run

Vector layers

Layer	Features	Total area (ha)	Note
buildings	96,788	n/a	

Vector layers made from rasters carry class_id, class_name, color_hex, area_ha, area_m2 (equal-area projection), pct_aoi, cx, cy, pipeline and period; layers a pipeline wrote itself (buildings, roads, streams, fire points) keep their own attributes. Coordinates are WGS 84 (EPSG:4326). The GeoPackage also holds the 'aoi' outline and, for raster-derived layers, a dissolved '<layer>_classes' summary.

References

- Microsoft. Global ML Building Footprints (github.com/microsoft/GlobalMLBuildingFootprints), Open Data Commons Open Database License.
- Sirko, W. et al. (2021). Continental-scale building detection from high resolution satellite imagery. arXiv:2107.12283. (Google Open Buildings v3.)
- OpenStreetMap contributors. Data (c) OpenStreetMap, available under the Open Database License (openstreetmap.org/copyright).